

TOKEN BALANCING, UPDATE SPARSITY, AND SUPERVISION DENSITY IN MULTI-TEACHER ON-POLICY DISTILLATION

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ABSTRACT

Multi-teacher on-policy distillation (MOPD) trains a single student using specialized teachers, but the effects of loss normalization, vocabulary supervision, and optimizer dynamics on parameter updates and task performance are not well understood. We study these effects with a Qwen3-1.7B student and four teachers for mathematics, code, instruction following, and science. First, averaging losses over tokens rather than responses changes gradient directions even when total loss weights are equal between domains. Nevertheless, three normalization rules differ by only 0.14 percentage points in final mean score under student-teacher top-64 intersection supervision with Adam. Second, after 500 updates, training changes about 97% of FP32 master weights but only 7.3–8.5% of stored BF16 weights. Teacher-specific Adam updates appear aligned largely because they use the same accumulated moment estimates: subtracting the update that Adam would take with no new gradient reduces their mean cosine similarity from above 0.83 to below 0.01 on domain-specific inputs. Third, the intersection of the student’s and teacher’s top-64 token sets contains over 99.9% of student probability mass on average at the evaluated checkpoints, and expanding supervision to the full vocabulary barely changes single-step update concentration. Yet we find no consistent performance advantage for top-64 over sampled-token supervision: at the joint-training endpoint, mathematics increases by 2.60 points while GPQA decreases by 1.77 points, with both paired 95% confidence intervals including zero. Thus sparse BF16 changes do not imply sparse optimization, and aligned Adam updates do not imply aligned teacher gradients. Neither statistic can replace per-task evaluation when choosing a distillation objective.

1 INTRODUCTION

Multi-teacher on-policy distillation (MOPD) is used in LLM post-training to transfer specialized capabilities into a single model (Xiaomi LLM-Core Team, 2026; NVIDIA, 2026; Kimi Team et al., 2026; DeepSeek-AI et al., 2026). The student generates responses, and a teacher assigned to each prompt domain supplies token-level supervision. The training objective is to improve several capabilities while preserving the student’s existing performance. Achieving this requires deciding how much each domain and response contributes to the loss, and how many next-token alternatives the teacher supervises.

Recent work establishes both the potential of MOPD and the sensitivity of its results to these choices. Ma et al. (2026) study capability integration with policy-gradient and top- k distillation objectives. Open-MOPD finds uneven transfer between domains and introduces domain token balancing, weighting by student-teacher performance gaps, and refreshed rewards (Gao et al., 2026). The Nemotron 3 Ultra report finds no advantage for top- k or full-vocabulary matching over sampled-token supervision in its preliminary experiments (NVIDIA, 2026, Section 3.3.5). Studies of OPD parameter changes also report concentration in a small set of stored weights (Yu et al., 2026; Shen et al., 2026). These findings leave open how the loss weights and vocabulary targets affect the gradients, optimizer updates, and task scores of a jointly trained student.

Three distinctions matter for answering this question. First, equal numbers of prompts per domain do not imply equal domain contributions to a token-averaged loss, because response lengths differ. Even after fixing each domain’s total loss weight, averaging over tokens or over responses gives different weights to long responses. Second, Adam updates depend on accumulated first- and second-moment estimates as well as the current gradient, and small FP32 updates may disappear when weights are stored in BF16. Sparse stored-weight changes or similar Adam update directions can therefore reflect the optimizer and numerical precision. Third, supervising more vocabulary tokens need not improve every task, even when the resulting parameter updates have similar concentration.

We study these distinctions through training runs and controlled gradient and optimizer comparisons with Qwen3-1.7B and four domain teachers. The main findings are:

1. **Different normalization rules produce distinct gradients but similar final top-64/Adam scores (Section 3).** With equal prompt counts, mathematics receives 44.05% of the token-averaged loss weight and instruction following 14.98%. Fixing domain weights does not remove response-length weighting: token and response averaging have a mean gradient cosine of 0.680. Applying the same Adam moment estimates raises their update cosine to 0.960. The three normalization rules finish within 0.14 mean-score points under top-64 supervision, although their learning curves differ; sampled-token/SGD runs have a larger 1.07-point spread.
2. **BF16 sparsity and Adam update alignment can obscure the current teacher gradient (Section 4).** Joint training changes about 97% of FP32 master weights but only 7.3–8.5% of stored BF16 weights. At the same student checkpoint, teacher-specific Adam updates have mean cosine similarity above 0.83 on their respective domain inputs. Removing the step that Adam would take using its accumulated moments with a zero current gradient reduces the cosine below 0.01. Thus aligned optimizer steps do not establish aligned current teacher gradients.
3. **More vocabulary supervision does not consistently improve task performance (Section 5).** Sampled-token, top-64, and full-vocabulary losses produce similarly concentrated single-step updates when the top-64 intersection already contains over 99.9% of student probability on average. In joint training, top-64 supervision scores 2.60 points higher on mathematics and 1.77 points lower on GPQA than sampled-token supervision at update 500; all paired task intervals include zero. Its mean advantage also shrinks from 1.69 points at update 250 to 0.29 points at update 500.

These results motivate evaluating supervision choices with per-task learning curves in addition to gradient and parameter statistics. Section 6 evaluates development-selected checkpoints on separate test sets: adding a per-domain retention constraint does not change any selected checkpoint in the ten tested trajectories.

2 PRELIMINARIES

Multi-teacher on-policy distillation (MOPD). For task domain d , the student generates a response $y \sim p_\theta(\cdot | x)$ to a prompt $x \sim \mathcal{D}_d$. A frozen teacher q_d assigned to that domain provides next-token probabilities at each student-generated prefix $h_t = (x, y_{<t})$. The objective depends on two choices: which vocabulary tokens enter the distillation loss at each prefix, and how token losses are averaged over responses and domains (Gao et al., 2026). For a rollout batch $\mathcal{B}$, let T_i be the number of valid tokens in response i and $\ell_i = T_i^{-1} \sum_t \ell_{i,t}$ its mean token loss, where the sum includes valid response positions only. The batch objective is

$$\mathcal{L}(\theta) = \sum_{i \in \mathcal{B}} w_i \ell_i(\theta), \quad w_i \geq 0, \quad \sum_{i \in \mathcal{B}} w_i = 1. \quad (1)$$

The token loss $\ell_{i,t}$ determines the distillation objective; w_i is the weight on response i ’s mean loss. As defined below, these response weights implement either token-level or sequence-level normalization, with or without explicit domain loss weights. Gradients treat the generated responses, valid-token masks, and weights w_i as fixed. Each training update uses fresh student-generated responses.

Sampled-token, top- k , and full-vocabulary supervision. These objectives use teacher probabilities for different numbers of vocabulary tokens at each prefix. Write $p_\theta(v) = p_\theta(v \mid h_t)$ and $q_d(v) = q_d(v \mid h_t)$. Full-vocabulary supervision minimizes the reverse KL divergence from the student to the teacher over vocabulary $\mathcal{V}$:

$$\ell_{\text{full}} = \sum_{v \in \mathcal{V}} p_\theta(v) \log \frac{p_\theta(v)}{q_d(v)}. \quad (2)$$

Sampled-token supervision uses a policy-gradient (PG) surrogate with the teacher–student log-probability ratio evaluated only at the generated token $a = y_t$:

$$\ell_{\text{PG}}(a) = -\text{sg}[\log q_d(a) - \log p_\theta(a)] \log p_\theta(a), \quad a \sim p_\theta. \quad (3)$$

Here sg denotes stop-gradient. At a fixed prefix, the expectation of the PG gradient over $a \sim p_\theta$ equals the full-vocabulary reverse-KL gradient (Appendix D).

Top- k supervision restricts the loss to a candidate set of likely next tokens. We use the intersection of the student and teacher top- k sets: let S and T contain their respective k most probable tokens, computed at rollout and held fixed during gradient computation, and let $I = S \cap T$. The loss is

$$\ell_{\text{top-}k} = - \sum_{v \in I} \text{sg} \left[\frac{p_\theta(v)}{\sum_{u \in S} p_\theta(u)} (\log q_d(v) - \log p_\theta(v)) \right] \log p_\theta(v). \quad (4)$$

The probabilities use the full-vocabulary softmax. The detached weights are normalized over the student set S before the sum is restricted to I ; an empty intersection gives zero loss. We denote $k = 16$ and $k = 64$ by I16 and I64, respectively. We use *supervision density* to mean the number of vocabulary tokens included in the loss at each prefix: one for PG, $|I| \leq k$ for intersection supervision, and $|\mathcal{V}|$ for full-vocabulary supervision. The model’s vocabulary size is unchanged.

Domain weighting and loss normalization. We compare three choices of w_i in Equation 1. Let $\mathcal{B}_d$ contain responses from domain d , and let $\lambda_d \geq 0$, with $\sum_d \lambda_d = 1$, denote the target domain loss weights. These weights are distinct from the number of prompts sampled per domain. We assume nonempty responses and at least one response per domain. For $i \in \mathcal{B}_d$,

$$w_i = \begin{cases} \frac{T_i}{\sum_{j \in \mathcal{B}} T_j}, & \text{global-token (GT),} \\ \frac{\lambda_d T_i}{\sum_{j \in \mathcal{B}_d} T_j}, & \text{domain-token (DT),} \\ \frac{\lambda_d}{|\mathcal{B}_d|}, & \text{domain-response (DR).} \end{cases} \quad (5)$$

GT averages over all valid tokens in the batch, so a domain’s total loss weight equals its fraction of the batch’s response tokens. DT first averages over tokens within each domain, then combines domain losses using λ_d . DR first averages over tokens within each response, then over responses within each domain, and finally combines domain losses using λ_d . Thus DT and DR use the same domain weights but different within-domain normalization: DT assigns longer responses proportionally more weight, whereas DR assigns equal weight to each response’s mean loss.

Cosine similarity between parameter updates. Let u_d be the parameter change from one optimizer step using domain d ’s mean loss. To compare domains, each step starts from an identical copy of the student weights and optimizer state, including Adam’s first- and second-moment estimates. For nonzero updates, define $C_{de} = (u_d^\top u_e) / (\|u_d\| \|u_e\|)$. Positive values indicate that the update vectors point in similar directions; negative values indicate opposing directions. Update cosine measures parameter-space geometry and does not by itself establish whether training improves or degrades a task’s performance.

3 LOSS NORMALIZATION AND TASK PERFORMANCE

3.1 EQUAL DOMAIN WEIGHTS DO NOT REMOVE RESPONSE-LENGTH WEIGHTING

Sampling equal numbers of prompts per domain does not give each domain equal weight in the training loss, because student response lengths differ by domain. With token-level normalization

over the full batch (GT), domains with longer responses receive more loss weight. Domain-token normalization (DT) fixes each domain’s weight to λ_d but still assigns longer responses more weight within that domain. Domain-response normalization (DR) also fixes the domain weights and gives equal weight to each response’s mean loss. Domain balancing and within-domain loss normalization are therefore separate design choices, even with a fixed prompt mixture.

This difference changes the gradient whenever response length correlates with the response gradient. Write $g_i = \nabla_{\theta} \ell_i$, and let $\bar{g}_d$ and $\bar{T}_d$ denote the mean response gradient and response length within domain d . The token-weighted domain gradient satisfies

$$\frac{\sum_{i \in \mathcal{B}_d} T_i g_i}{\sum_{i \in \mathcal{B}_d} T_i} = \bar{g}_d + \frac{\text{Cov}_d(T, g)}{\bar{T}_d}. \quad (6)$$

The covariance term measures the effect of weighting response gradients by length. DT retains this term; DR uses $\bar{g}_d$ alone. Multiplying each domain’s GT loss contribution by its target weight λ_d divided by its observed token share gives DT exactly, as in Open-MOPD (Gao et al., 2026, Section 4.1, Eq. (8)). Thus correcting unequal domain token counts still leaves a choice between token-level and sequence-level normalization within each domain. Appendix B gives the derivation.

3.2 SIMILAR I64 ENDPOINT SCORES DESPITE DIFFERENT LOSS WEIGHTS

Training and evaluation. We train a Qwen3-1.7B student with four frozen teachers for mathematics, code, instruction following, and science. The three I64 configurations differ only in GT, DT, or DR loss normalization. They use the same initialization, teachers, prompt schedule, and optimizer settings, with 16 prompts per domain at each update. We evaluate MATH-500, a fixed LiveCodeBench subset, IFBench, and GPQA Diamond through update 500 and report their equally weighted mean. Appendix A gives the training and scoring protocol.

Correcting a threefold loss-weight imbalance changes little at the I64 endpoint. Mathematics and instruction following each receive 25% of prompts but account for 44.05% and 14.98% of GT’s valid response tokens in the first 50 batches. GT therefore assigns mathematics nearly three times as much loss weight as instruction following, whereas DT and DR fix both at 25% (Figure 1a). Despite this difference, the update-500 four-domain scores are $38.58 \pm 0.75\%$ (DR), $38.71 \pm 0.80\%$ (DT), and $38.64 \pm 0.80\%$ (GT), reported as mean $\pm$ standard deviation over three training seeds. The 0.14-percentage-point range between means is smaller than the 0.75–0.80-point seed standard deviations. The preferred rule varies with the task and checkpoint: all three improve mathematics and instruction following toward teacher performance, retain a code performance gap, and start near the science teacher’s GPQA score (Figure 1). Equal domain loss weights therefore do not consistently improve the final mean score in this I64/Adam setting.

Normalization changes PG/SGD endpoint scores by 1.07 points. In the seed-42 endpoint comparison, the four-domain score range between normalization rules is 0.33 points for PG/Adam and 1.07 points for PG/SGD, compared with 0.14 points for I64/Adam (Table 1). PG/SGD favors DR over GT by 1.07 points, while PG/Adam favors GT over DR by 0.07 points. The small differences under I64/Adam therefore do not establish that normalization is unimportant for other training configurations. All ten configurations improve the initial four-domain mean from 33.93% to 38.03–39.46%. These results use one training seed, unlike the three-seed I64 summary above. The SGD runs also use a different learning rate and deployment, so their higher scores do not isolate the effect of the optimizer or establish a training-efficiency advantage. Appendix H reports the separate test-set evaluation.

3.3 ADAM REDUCES DIRECTIONAL DIFFERENCES BETWEEN NORMALIZATION RULES

Different gradients produce more similar Adam updates. I64 gradient cosines average 0.680 (DR–DT), 0.483 (DR–GT), and 0.822 (DT–GT) on six fixed response batches at joint-PG checkpoints 100 and 250. These comparisons hold the student weights, responses, token masks, and teacher scores fixed, so the directional differences come from loss normalization. In particular, the DR–DT comparison shows that equal domain weights do not eliminate the effect of response-length weighting. Applying each gradient from an identical copy of the saved Adam first- and second-moment estimates raises the corresponding FP32 update cosines to 0.960, 0.939, and 0.979. The

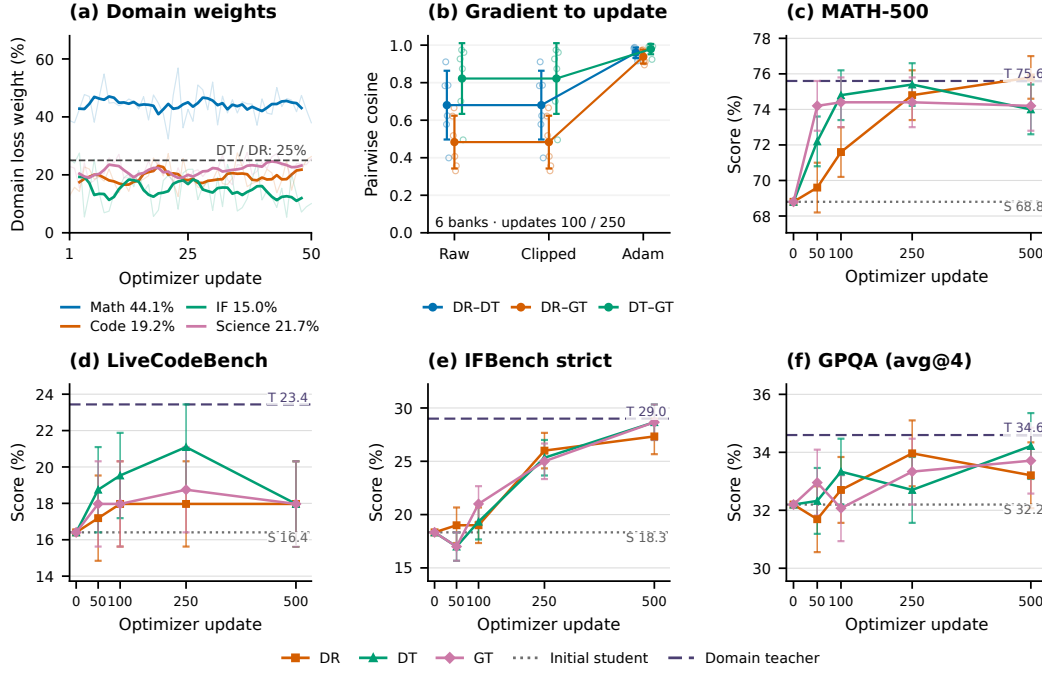


Figure 1: **Loss averaging changes gradient directions and task performance; Adam increases update alignment.** (a) GT domain token shares: individual batches and five-batch averages. (b) Cosines before/after clipping and after Adam (batch means $\pm$ SD). (c–f) I64 task scores (three training seeds, mean $\pm$ SD); S/T mark the initial student/teacher.

Table 1: **Every configuration improves the four-domain mean, with different domain-level trade-offs.** Joint-training endpoint scores (%). Mean weights domains equally.

Configuration	MATH-500	Code	IF	GPQA	Mean
Initial student	68.80	16.41	18.33	32.20	33.93
<i>Adam</i>					
PG-DR	73.20	17.97	27.00	34.97	38.29
PG-DT	73.20	17.97	27.00	33.96	38.03
PG-GT	76.20	17.97	27.33	31.94	38.36
I16-DR	75.40	18.75	27.00	33.46	38.65
I64-DR	75.80	17.97	27.33	33.21	38.58
I64-DT	74.00	17.97	28.67	34.22	38.71
I64-GT	74.20	17.97	28.67	33.71	38.64
<i>SGD</i>					
PG-DR	76.40	19.53	28.33	33.59	39.46
PG-DT	75.20	20.31	28.33	32.70	39.14
PG-GT	75.00	19.53	25.33	33.71	38.39

resulting parameter updates are therefore much more similar in direction than the gradients used to compute them.

Gradient clipping changes magnitude, not direction. Global norm clipping leaves all three pairwise gradient cosines unchanged. Its frequency nevertheless differs sharply during training: DR is clipped on all 500 updates and retains a median 9.3% of the gradient norm, whereas DT and GT are clipped on 46.2% and 15.0% of updates, respectively (Figure 2a). Clipping alone therefore cannot explain the increase in update cosine under Adam. Resetting Adam’s moment estimates also changes



Figure 2: **DR triggers clipping at every update; Adam moments change step norms and one-step loss reductions.** (a) Retained gradient fraction; labels give clipping rates. (b) FP32 step norms under moment resets. (c) Held-out reverse-KL changes (lower is better). Open points are cached batches; filled points and error bars show mean $\pm$ SD.

the update norms (Figure 2b), showing that the saved optimizer state affects update magnitude as well as direction.

One-step loss changes also depend on the optimizer state. After one Adam step from the same saved moment estimates, the three normalization rules produce similar changes in held-out reverse KL relative to the variation between response batches (Figure 2c). When SGD steps are rescaled to match each rule’s Adam update norm, the differences between rules are larger: all three reduce mean held-out KL at update 100 but increase it at update 250. These measurements concern the immediate effect of one update on fixed held-out responses. They do not account for the complete training trajectories, in which each student generates different responses and accumulates different Adam moments. Appendix B.1 gives the one-step protocol and gradient comparisons; Figure 6 reports paired task-score contrasts.

4 PARAMETER-UPDATE SPARSITY AND TEACHER ALIGNMENT

The joint student changes only a small fraction of its BF16 weights, even though the optimizer changes most FP32 master weights. In addition, updates computed from different teachers can point in similar directions because they use the same accumulated Adam moments. These results qualify the interpretation of concentrated parameter changes reported for distillation (Yu et al., 2026): both numerical precision and the optimizer state affect which changes are observed.

4.1 SPARSE BF16 CHANGES MASK WIDESPREAD FP32 UPDATES

Write $\Delta = \theta - \theta_0$ for cumulative changes from initialization and $u = \theta_{\text{after}} - \theta_{\text{before}}$ for an individual optimizer step. For either change vector z with P coordinates, we report the fraction of parameters whose absolute change exceeds threshold τ :

$$f_\tau(z) = \frac{1}{P} \sum_{j=1}^P \mathbf{1}\{|z_j| > \tau\}. \quad (7)$$

We also report the smallest coordinate fraction accounting for 90% of the squared change, which measures concentration without an absolute threshold. We compute both statistics for the optimizer’s FP32 master weights and the model’s stored BF16 weights. These are parameter-change measurements, distinct from raw gradients before clipping and the optimizer transformation.

The joint PG and I64-DR runs use the same training settings, including equal domain weights and equal response weights within each domain (Appendix A). After 500 updates, 7.30% of PG’s BF16 weights differ from initialization, and 3.79% of coordinates account for 90% of the squared change. The corresponding I64 fractions are 8.46% and 4.38% (Figure 3a). Both runs therefore concentrate their cumulative BF16 changes in a small fraction of parameters.

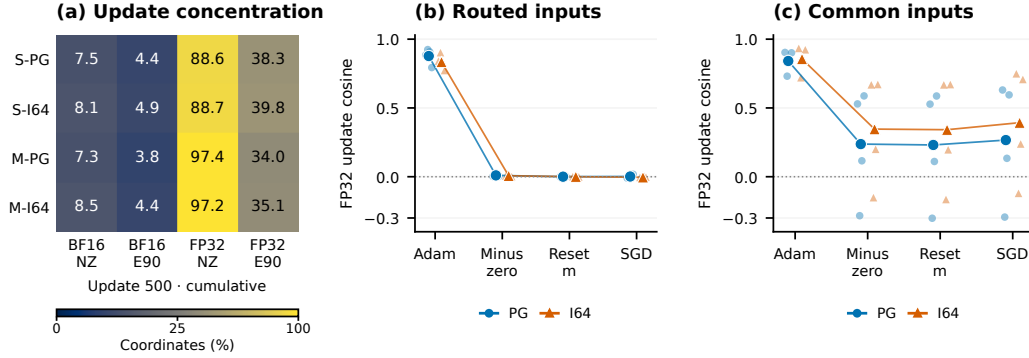


Figure 3: **BF16 parameter changes are sparse; accumulated Adam momentum explains most update alignment on domain-specific inputs.** (a) Nonzero (NZ) and 90%-energy (E90) coordinate fractions; S/M denote single-teacher/joint training. (b,c) FP32 update cosines on domain-specific/shared inputs. “Minus zero” subtracts the step from Adam with the current gradient set to zero; small/large points show batch/overall means.

In FP32, however, 97.37% of PG’s master weights and 97.17% of I64’s differ from initialization; 34.03% and 35.14% of coordinates account for 90% of their squared changes. The individual optimizer step at update 500 changes 86.80% and 87.05% of FP32 coordinates, respectively. Most parameters thus receive FP32 updates, while many changes disappear when represented in BF16. Sparse differences between BF16 checkpoints do not imply that the optimizer updates only a small subset of parameters. Appendix Figure 10 gives the trajectories, and Appendix C reports threshold and optimizer-state controls.

4.2 ADAM MOMENTS ALIGN UPDATES FROM DIFFERENT TEACHERS

We compute a separate optimizer step for each teacher using identical copies of the joint student and its saved Adam state, including the first and second moments. For teacher d , let $S_d(\rho)$ contain the largest-magnitude ρ fraction of its parameter updates. We compare these coordinate sets using Jaccard overlap:

$$J(d, e) = \frac{|S_d(\rho) \cap S_e(\rho)|}{|S_d(\rho) \cup S_e(\rho)|}. \quad (8)$$

Equal selection sizes make teacher pairs comparable. Jaccard overlap measures whether the largest updates occur at the same coordinates, whereas cosine similarity also accounts for update signs and magnitudes. Teachers can update the same coordinates in opposing directions, as distinguished in the support-overlap analysis of Yu et al. (2026).

At the joint PG checkpoint after 500 updates, we compare teachers given the same prefixes (common inputs) and teachers given prefixes from their assigned domains (routed inputs). Each result averages the six teacher pairs within each of four sampled response batches, then averages the batches; these repetitions do not measure training-seed variation. With routed inputs and the saved Adam moments, the mean pairwise cosine between FP32 updates is 0.8774 for PG and 0.8347 for I64. Resetting the first moment while retaining the second, or replacing Adam with SGD scaled to the same step norm, reduces both mean cosines to near zero (Figure 3b). The accumulated first moment therefore accounts for much of the alignment between these teacher-conditioned steps. Appendix H.5 relates coordinate overlap to update cosine and teacher-distribution similarity; Appendix C.2 controls for input choice when comparing raw gradients.

4.3 SUBTRACTING THE ZERO-GRADIENT STEP REMOVES MOST UPDATE ALIGNMENT

Adam can change parameters even when the current gradient is zero: its first moment retains gradients from earlier updates. To measure how a teacher’s current gradient changes the proposed step, we subtract this zero-gradient baseline. Let $U(g; H)$ be the parameter update computed from gradient g using saved state H , which includes the first and second moments and the step count. For

teacher d , we compute

$$\delta U_d = U(g_d; H) - U(0; H). \quad (9)$$

Both steps start from identical parameters and optimizer state. Thus δU_d measures the change in the Adam step caused by supplying g_d instead of zero. It still depends on the saved moments, and Adam’s nonlinear transformation prevents interpreting it as an additive teacher contribution.

With routed inputs, the mean pairwise cosines fall from 0.8774/0.8347 for the complete PG/I64 steps to 0.0098/0.0067 for δU_d (Figure 3b). With common inputs, they fall from 0.8415/0.8562 to 0.2373/0.3462 (panel c). After subtraction, teachers given their respective domain prefixes therefore produce update differences whose pairwise cosine is nearly zero on average, despite the strong alignment of their complete Adam steps. Similar complete steps need not reflect similar directions induced by current teacher gradients. Neither coordinate overlap nor update cosine establishes whether one teacher improves or harms another domain’s performance; that requires per-domain loss or capability evaluation.

5 VOCABULARY SUPERVISION: PARAMETER UPDATES AND TASK PERFORMANCE

Supervising more vocabulary tokens per prefix does not necessarily make parameter updates less concentrated. Consistent with the concentrated checkpoint changes observed under dense distillation losses (Yu et al., 2026), we find nearly identical one-step update concentration for sampled-token, top- k , and full-vocabulary losses when applied to the same student and Adam state. End-to-end training nevertheless produces different task scores: I64 ends with higher mathematics accuracy and lower GPQA accuracy than PG. We examine both measurements to distinguish immediate optimizer behavior from the performance of the trained student.

5.1 VOCABULARY SUPPORT AND EXPERIMENTAL CONTROLS

PG supervises one sampled token at each prefix; I64 supervises the intersection of the student’s and teacher’s top-64 token sets (Equations 3 and 4). We compare S-PG and S-I64 trained with the mathematics teacher, and M-PG and M-I64-DR trained jointly with four domain teachers. Within each setting, the runs use the same initialization, teachers, domain-response averaging, optimizer settings, and prompt schedule. The single-teacher runs use 64 mathematics responses per update; joint runs use 16 per domain. We compare PG and I64 after the same number of optimizer updates. Joint I16 training uses the same intersection objective and averaging rule, with $k = 16$. These comparisons use seed 42.

The one-step comparison evaluates PG, I64, teacher-top-64 supervision (T64), and full-vocabulary reverse KL at four joint-training checkpoints: PG and I64-DR after 100 and 500 updates. T64 uses the teacher’s top-64 set without requiring membership in the student’s top-64 set, with the gradient correction defined in Appendix D. At each checkpoint, all four losses use the same 96 sampled prefixes, grouped into four batches, with identical prefix weights and saved Adam moments. We measure the fraction of changed parameters and the fraction of squared step norm carried by the largest 1% of coordinates.

5.2 UPDATE CONCENTRATION

More supervised tokens leave one-step concentration nearly unchanged. At M-PG/100, each of the four losses produces an optimizer step that changes about 88.0% of FP32 master weights (Table 6). After conversion to BF16, only 0.149% of parameters change for PG and 0.150% for I64, T64, and full vocabulary. The largest 1% of coordinates account for about 8.8% of the FP32 squared step norm and all of the BF16 squared step norm. BF16 nonzero fractions are also similar between losses at the other three checkpoints (Figure 4a). Adam’s accumulated moments contribute to these changes: even with a zero current gradient, its step changes 0.140% of BF16 parameters at M-PG/100.

At each of the four checkpoints, the actual I64 intersection retains more than 99.9% of student probability mass on average; the minimum over all 384 prefixes is 96.55%. Thus I64 already includes nearly all student probability mass in this comparison, despite supervising at most 64 vocabulary

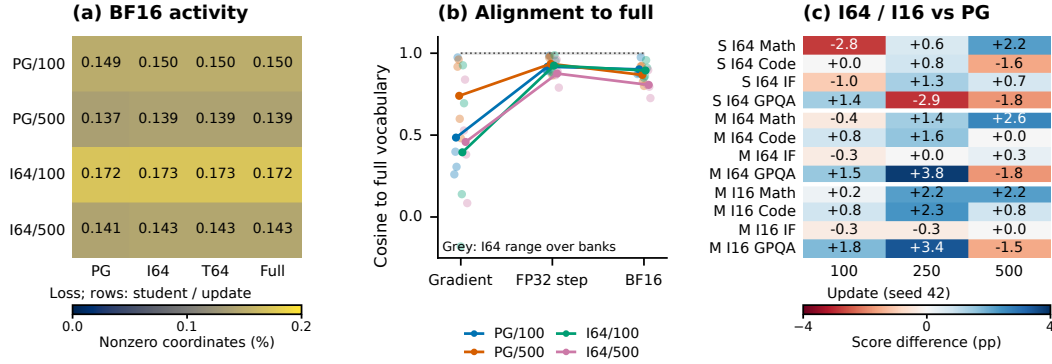


Figure 4: **I64 better approximates the full-vocabulary gradient, with similar BF16 update sparsity and mixed task gains.** (a) Nonzero BF16 changes (%). (b) Cosine to the full-vocabulary gradient or update; shading shows the I64 range. (c) I64/I16 minus PG scores (percentage points); S/M denote single-teacher/joint training.

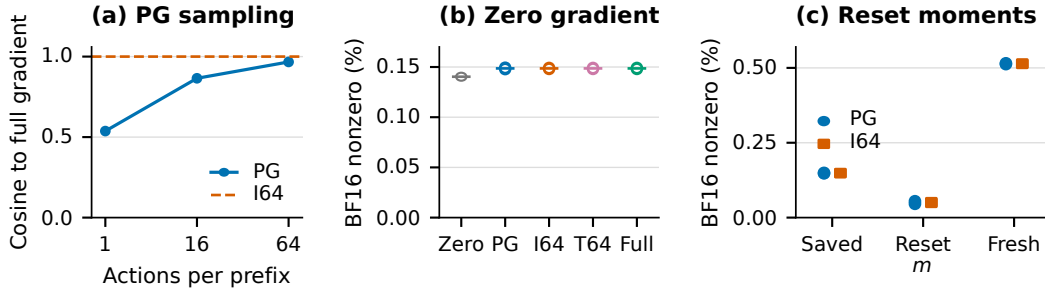


Figure 5: **More token samples increase PG gradient cosine to full-vocabulary KL; Adam momentum alone changes nearly as many BF16 parameters as PG.** (a) Gradient cosine to full-vocabulary KL. (b,c) Nonzero BF16 changes with a zero current gradient or reset Adam moments. Points denote cached batches; bars show means.

tokens. These measurements do not determine what happens when top- k sets omit substantial probability mass. Table 10 reports the intersection size, probability mass, and retained weight ratio by checkpoint; Appendix C provides threshold and optimizer-state controls.

The raw gradients still differ even when the resulting optimizer steps have similar concentration. In a separate experiment with fixed prefixes, averaging 1, 16, and 64 PG token samples raises mean gradient cosine with the full-vocabulary gradient from 0.537 to 0.865 to 0.967 (Figure 5a). Finite token sampling therefore explains part of the gradient difference between PG and full-vocabulary supervision. Separate zero-gradient and moment-reset controls quantify how the saved Adam moments affect the number of BF16 weights that change (panels b,c).

Similar single-step concentration does not imply similar cumulative changes. After 500 joint-training updates, 8.46% of I64’s BF16 weights differ from initialization, compared with 7.30% for PG (Section 4; Figure 10). The one-step experiment fixes the student, prefixes, and Adam moments. During end-to-end training, each loss instead changes the student, its subsequent sampled responses, and the accumulated optimizer moments. The similarity of one-step concentration therefore does not extend to the number of parameters changed over training.

5.3 CAPABILITY UNDER SINGLE-TEACHER AND JOINT TRAINING

I64’s mathematics advantage emerges late and does not extend to every task. In single-teacher training, PG leads I64 on mathematics by 2.8 percentage points after 100 updates, but I64 leads PG

by 2.2 points after 500 updates (Figure 4c). At update 500, I64 also scores 0.67 points higher on instruction following, but 1.56 points lower on code and 1.77 points lower on GPQA. Appendix Table 2 gives the scores, and Table 4 gives the paired uncertainty intervals.

Joint training shows the same endpoint trade-off between mathematics and GPQA: I64-minus-PG differences are +2.60 points on mathematics, 0.00 on code, +0.33 on instruction following, and -1.77 on GPQA (Table 1). All four nominal paired 95% intervals include zero, so these point estimates do not establish a consistent task-level improvement (Table 4). The ranking also changes during training. At update 250, I64 scores higher on mathematics, code, and GPQA and ties PG on instruction following; its four-domain mean advantage then shrinks from 1.69 to 0.29 points by update 500.

Top-64 supervision does not consistently outperform top-16. Joint I16 and I64 have endpoint four-domain means of 38.65% and 38.58%, respectively (Table 1). On the separate test sets, their means are again close, at 44.56% and 44.46%, but I64 scores 1.96 points higher on mathematics and 1.17 points higher on science, while I16 scores 2.35 points higher on code and 1.17 points higher on instruction following (Table 8). These estimates establish neither statistical equivalence nor a general advantage for either candidate-set size. Changing k changes both the supervised token set and the normalization of token weights in Equation 4; it does not isolate token count alone. I16 is evaluated in end-to-end training, without a corresponding one-step geometry measurement.

6 PRACTICAL IMPLICATIONS AND CHECKPOINT SELECTION

The experiments identify two reasons to assess MOPD with per-task evaluations: loss normalization changes the relative contribution of domains and responses, while optimizer moments and weight precision can hide differences between current gradients.

Choose domain and response weights explicitly. Specify both the domain loss weights and whether losses are averaged over tokens or responses; equal prompt counts do not define either choice. The PG/I16/I64 comparisons likewise show no consistent benefit from increasing the number of supervised vocabulary tokens. Choose this number using task scores and measured training cost, and report the probability mass covered by the candidate set. Our runs do not support a compute-efficiency ranking because their costs were not matched.

Separate the current gradient from Adam moments and BF16 rounding. Measure raw gradients, FP32 optimizer updates, and BF16 stored-weight changes separately. For teacher comparisons, start each proposed update from identical student parameters and Adam moment estimates. The zero-gradient reference measures the step driven by the accumulated moments without a new teacher gradient. Its subtraction reveals whether similar complete updates also have similar responses to the current teacher signals. Overlapping changed coordinates or high update cosine alone cannot establish that training with one teacher preserves another task’s performance.

A retention constraint does not improve checkpoint selection in these runs. We test whether ruling out checkpoints with a large task regression improves selection by the mean development score. Let $s_d(\theta)$ be the development score for domain d , with higher values better. Eligible checkpoints satisfy

$$s_d(\theta) \geq s_d(\theta_0) - \epsilon_d \quad \text{for every domain } d, \quad (10)$$

where $\epsilon_d \geq 0$ is a prespecified tolerance relative to the initial student. We select the eligible checkpoint with the highest equal-domain mean, keeping the initial student if none improves that mean by at least one percentage point. We compare this rule with maximizing the same mean without the domain constraint and with using the final checkpoint. The tolerance is two percentage points per domain, fixed before the additional test results (Appendix H.2).

The additional test contains 256 separately selected questions per domain and uses a distinct evaluation protocol (Appendix H.1). The retention-constrained and unconstrained rules select the same checkpoint in all ten trajectories: update 500 in nine cases and update 250 for PG-DT/Adam. For PG-DT/Adam, the selected checkpoint scores 43.14% on the additional test, below the final checkpoint’s 43.51%. Thus neither the constraint nor the earlier selected checkpoint provides an observed

test-score benefit here. Method rankings also depend on the evaluation set: PG-DR/SGD exceeds Adam by 1.18 mean points on the original benchmarks but is 0.07 points lower on the additional test. Because the development benchmarks had already been inspected, this checkpoint-selection analysis is exploratory.

7 RELATED WORK

Loss normalization in LLM reinforcement learning. DAPO replaces response-level averaging with token-level averaging to change the contribution of long responses (Yu et al., 2025). Dr. GRPO analyzes the optimization bias from response-length normalization and scaling advantages by the within-group reward standard deviation (Liu et al., 2025). Our analysis concerns a distinct choice in multi-domain distillation: fixing each domain’s total loss weight still leaves token-level and response-level averaging within that domain. We measure how this choice changes gradients, Adam updates, and task performance.

Multi-teacher on-policy distillation. MOPD combines specialized teachers through supervision on student-generated responses (Ma et al., 2026). Open-MOPD addresses uneven capability transfer with domain token balancing, performance-gap weighting, and reward refresh (Gao et al., 2026). For the same batch and target domain weights, its token-share correction is equivalent to our domain-token averaging. We further separate this domain correction from response-length weighting within each domain.

Parameter changes during post-training. Mukherjee et al. (2025) investigate small parameter subsets that support reinforcement-learning fine-tuning. Yu et al. (2026) find concentrated checkpoint changes under dense distillation supervision and overlapping changes in students trained separately with different teachers. We instead compare teacher-specific updates from identical copies of a jointly trained student and its Adam state, and distinguish FP32 optimizer updates from BF16 stored-weight changes. Shen et al. (2026) study OPD parameter geometry and select response positions for learning. Our vocabulary-supervision comparison varies the candidate tokens scored at each position and evaluates both parameter updates and task scores.

8 DISCUSSION

The main empirical lesson is that parameter statistics can hide differences that matter for interpreting MOPD. Adam’s accumulated moments make distinct current gradients produce similar updates, and BF16 storage makes dense FP32 parameter changes appear sparse. Neither observation establishes that teachers train disjoint parameter subsets or improve tasks without interference. Conversely, similar single-step concentration under different vocabulary losses does not imply the same task scores after training. In our runs, loss normalization has little effect on the final top-64/Adam mean, whereas the relative performance of sampled-token and top- k supervision changes with the task and checkpoint.

The scope of these findings is limited to one Qwen3-1.7B student and four teachers fine-tuned from the same initial model. The normalization curves summarize seeds 42–44; the expanded objective/optimizer comparison and additional test use seed 42. Repeated response batches measure diagnostic sampling variation, while question-bootstrap intervals measure evaluation uncertainty for fixed checkpoints; neither estimates variation between independent training runs. The top-64 intersections in the single-step comparisons already retain almost all student probability, so the results do not characterize low-probability vocabulary tails or more dissimilar teachers. Appendix H.4 removes four known overlapping mathematics questions as a sensitivity check, rather than an exhaustive contamination audit. Broader conclusions about training stability or compute efficiency require repeated runs with matched compute budgets.

REPRODUCIBILITY STATEMENT

The accompanying evidence bundle contains figure and table values, configuration records, source hashes, scoring audits, and CPU plotting scripts. Appendix A gives the training data sources, eval-

uation protocol, evaluated checkpoints, and uncertainty estimates. Appendix C specifies parameter measurements and single-step optimizer controls; Appendix H describes the additional test set, checkpoint-selection rules, and sensitivity to known training–evaluation overlap.

ETHICS STATEMENT

The experiments use public model families and benchmark datasets to study optimization and capability transfer. We report standard benchmark scores with the stated evaluation denominators. Generative AI assisted manuscript editing and analysis code; the authors are responsible for the reported measurements and their interpretation.

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A EXPERIMENTAL PROTOCOL AND CAPABILITY RESULTS

The experiments use a shared student initialization and a pool of four domain-specific teachers. We compare loss averaging rules, optimizer updates from different teachers, and the number of vocabulary tokens included in the distillation loss. The ten joint-training configurations in Table 1 include PG-DR and I64-DR/DT/GT. Mathematics-only PG/I64 runs provide the single-teacher comparison. All checkpoint axes count optimizer updates, and each trajectory comparison uses checkpoints completed for every included branch.

A.1 MODEL, DATA, AND TRAINING PROTOCOL

The student is Qwen3-1.7B (Yang et al., 2025). The four frozen teachers were trained by reinforcement learning from the same initial model, specializing in mathematics, code, instruction following, and science, respectively. Student and teachers share the tokenizer and native chat template with thinking disabled. The experiment records identify the model revisions and weight files.

AdamW configurations use learning rate 2.5×10^{-7} with $(\beta_1, \beta_2) = (0.9, 0.98)$, $\epsilon = 10^{-8}$, no weight decay or schedule, and global norm-1 clipping. Training sampling uses temperature 1 and top- $p = 1$, with a 2,048-token prompt cap and 4,096-token response cap. One fresh response is sampled per prompt. Training uses seeds 42–44 for the I64 GT/DT/DR normalization capability summaries and seed 42 for the other comparisons. Each optimizer update aggregates 64 responses; domain allocation is specified below.

The multi-teacher configurations allocate 16 prompts to each domain per update and use target domain weights $1/4$. Single-teacher configurations allocate all 64 prompts to mathematics. Prompts follow a seeded shuffled schedule, repeated after exhaustion. I64 uses the top- k intersection objective in Equation 4 with $k = 64$ from initialization; I16 uses $k = 16$. The supplementary PG/SGD branches use a learning rate of 0.0025 and each complete 500 updates. Hardware and deployment differ from the AdamW runs, so the capability comparison does not establish a runtime ranking.

Training prompts come from the Nemotron-3-Nano-RL-Training-Blend: the DAPO and Skywork mathematics subsets, competition coding with unit tests, instruction following, and STEM multiple-choice questions. The frozen derived four-domain release contains the processed prompts and scoring metadata. A seeded shuffle reserves separate 64-record diagnostic and held-out splits per domain; the remaining training splits contain 21,928 mathematics, 19,041 code, 16,447 instruction-following, and 19,542 science records.

A.2 EVALUATION AND SCORING

Each capability suite contains MATH-500 (500 questions from the MATH benchmark (Hendrycks et al., 2021); dataset release), a fixed 128-question LiveCodeBench v6 subset (Jain et al., 2024), IFBench strict (300 questions) (Pyatkin et al., 2025), and GPQA Diamond (198 questions) (Rein et al., 2023). Mathematics, code, and instruction following use one greedy response per question. GPQA uses four responses per question at temperature 1 and top- $p = 1$. Its score is the mean correctness over all four responses, then over questions: average@4 accuracy. Evaluation permits up to 32,768 response tokens, bounded by the remaining configured 32,768-token context. Each complete suite contains 1,720 responses.

The expanded development grid comprises updates 50/100/250/500 for seven Adam trajectories and 100/250/500 for three SGD trajectories, plus the initial student evaluation. Appendix H.1 describes additional test sets for the same four domains. Table 2 supplements the joint endpoints in Table 1 with single-teacher scores and intermediate PG/I64 checkpoints.

The four RL teachers were evaluated separately on their respective domain benchmarks using the same prompt and decoding protocol (Table 3). These scores quantify each teacher’s performance on its target domain.

A.3 PAIRED COMPARISONS AND EXPERIMENT RECORDS

We verify question correspondence and bootstrap the per-question score differences 10,000 times using seed 1042; each GPQA question retains its four-response cluster. Figure 6 shows percentile

Table 2: **PG/I64 performance depends on the task and checkpoint.** Seed-42 scores (%): S is mathematics-only training and M is joint training; both use DR. Joint endpoints appear in Table 1.

Configuration	Updates	Math	Code	IF	GPQA
S-PG	100	77.40	18.75	18.67	31.82
S-PG	250	73.80	20.31	17.00	35.35
S-PG	500	73.80	19.53	18.00	31.82
S-I64	100	74.60	18.75	17.67	33.21
S-I64	250	74.40	21.09	18.33	32.45
S-I64	500	76.00	17.97	18.67	30.05
M-PG	50	69.60	16.41	18.67	28.03
M-PG	100	72.00	17.19	19.33	31.19
M-PG	250	73.40	16.41	26.00	30.18
M-I64-DR	50	69.60	17.19	19.00	31.69
M-I64-DR	100	71.60	17.97	19.00	32.70
M-I64-DR	250	74.80	17.97	26.00	33.96

Table 3: **Reference scores for the four domain-specific teachers (%)**. Science uses GPQA average@4.

Teacher / target domain	Responses	Score (%)	Truncation (%)
Math	500	75.60	6.80
Code	128	23.44	24.22
If	300	29.00	7.33
Science	792	34.60	0.51

95% intervals. These are nominal, unadjusted intervals over evaluation questions, for the evaluated checkpoints. Training-seed variation is a separate source of uncertainty. We use the same procedure for comparisons with the initial student.

The experiment archive retains scores, question-level results or verified run summaries, model and data identifiers, and analysis scripts. The initial and expanded evaluations reuse some checkpoints and should not be counted as independent runs. We distinguish uncertainty from training seeds, evaluation questions, and repeated sampling of cached response batches (called *banks* in the figures).

B LENGTH-WEIGHTING DERIVATION

Fix a rollout batch, its valid-token masks, and domain loss weights λ_d . The PG log-ratio coefficients are treated as constants during backpropagation. For $n_d = |\mathcal{B}_d|$, let $\bar{T}_d = n_d^{-1} \sum_{i \in \mathcal{B}_d} T_i$ and $\bar{g}_d = n_d^{-1} \sum_{i \in \mathcal{B}_d} g_i$. Define the empirical covariance using the same $1/n_d$ normalization:

$$\text{Cov}_d(T, g) = \frac{1}{n_d} \sum_{i \in \mathcal{B}_d} (T_i - \bar{T}_d)(g_i - \bar{g}_d). \quad (11)$$

Expanding gives $n_d^{-1} \sum_i T_i g_i - \bar{T}_d \bar{g}_d$. Dividing by $\bar{T}_d$ gives Equation 6. GT combines the resulting domain gradients using observed token shares; DT combines them using λ_d ; DR combines $\bar{g}_d$ using λ_d . Multiplying each domain’s globally averaged terms by its target weight divided by its observed token share therefore gives DT exactly.

The derivation is a finite-batch weighted-average identity. Response sampling, masks, and lengths are held fixed. Its population analogue involves an expectation of a ratio; replacing that expectation by a ratio of expectations requires additional assumptions or a large-batch limit.

B.1 FIXED-BATCH GRADIENT COMPARISON

At M-PG checkpoints 100 and 250, three cached response batches (seeds 1042, 1043, and 1044) each contain eight responses per domain. Responses use the training prompt format and 4,096-token response cap. For each batch, we compute each response’s mean I64 gradient once at the

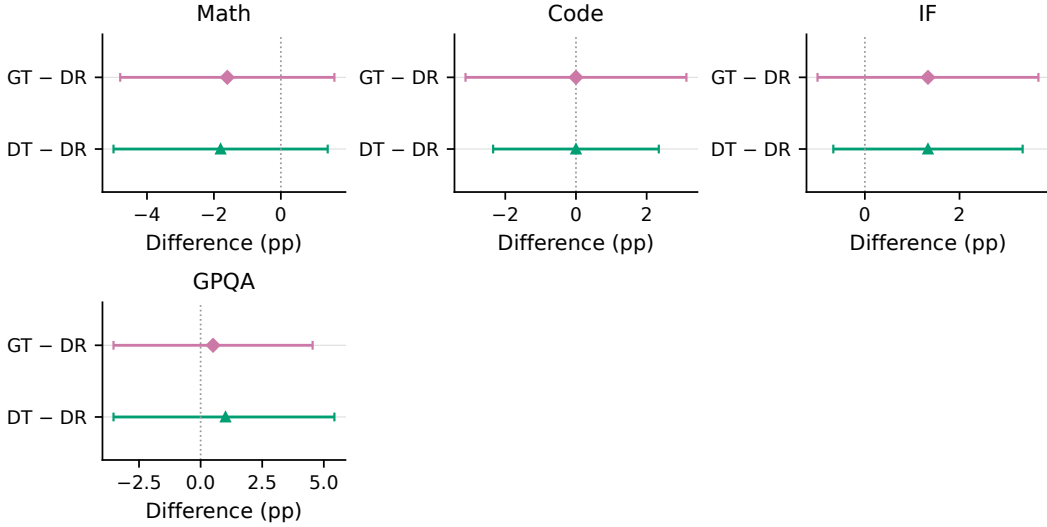


Figure 6: **I64 averaging rules produce task-dependent score differences.** Update-500 differences from DR with nominal 95% paired question-bootstrap intervals.

Table 4: **All eight I64-minus-PG intervals include zero.** Update-500 score differences and nominal 95% paired question-bootstrap intervals (percentage points).

Setting	Domain	I64 - PG (pp)	95% interval (pp)
Single	Math	+2.20	[-0.80, +5.00]
Single	Code	-1.56	[-3.91, +0.00]
Single	IF	+0.67	[-1.67, +3.00]
Single	GPQA	-1.77	[-5.30, +1.77]
Joint	Math	+2.60	[-0.40, +5.60]
Joint	Code	+0.00	[-2.34, +2.34]
Joint	IF	+0.33	[-2.00, +2.67]
Joint	GPQA	-1.77	[-5.81, +2.40]

saved BF16 model values using FP32 computation, then aggregate those gradients under DR, DT, and GT. All three averaging rules use the same responses, token masks, teacher scores, and model coordinates. Each comparison starts from the same student parameters and Adam first and second moments.

Table 5 reports every pairwise raw-gradient cosine. The domain-wise covariance decomposition is checked in FP64 for all 24 batch-domain groups; the largest relative residual is 1.53×10^{-15} . This numerical check verifies the aggregation identity. The measured cosine differences show that response length and gradient direction are correlated in these batches.

The optimizer interventions in Figure 2 reuse these six batches. Each aggregate gradient is globally clipped at norm 1 and applied to the same saved FP32 Adam state for that batch. The saved-state branch retains both moments; the reset- m branch zeros the first moment while retaining the second moment and optimizer step counter; reset- (m, v) zeros both moments while retaining the step counter. The momentum-free SGD control follows the negative gradient and is rescaled separately for DR, DT, and GT to match the FP32 Adam step norm for that averaging rule. After each FP32 update, the model parameters are cast to BF16 and evaluated on one held-out response per domain. Evaluation uses full-vocabulary reverse KL; the experiment archive also retains the I64 surrogate used for training. Each comparison applies a single optimizer step to a checkpoint copy.

Table 5: **Equal domain weights do not imply equal gradient directions.** Raw-gradient cosines on identical cached responses. DR and DT use equal domain loss weights; only DT weights responses by length.

Updates	Batch	DR-DT	DR-GT	DT-GT
100	1	0.7838	0.6219	0.9266
100	2	0.9110	0.5180	0.7010
100	3	0.5780	0.6659	0.9745
250	1	0.6240	0.4072	0.8742
250	2	0.7864	0.3302	0.4953
250	3	0.3990	0.3569	0.9612

C PARAMETER AND OPTIMIZER MEASUREMENTS

C.1 COORDINATES AND RECORDED QUANTITIES

Training measurements separately record gradients before and after clipping, FP32 optimizer steps, and cumulative parameter changes in the FP32 master weights and BF16 model weights. Cumulative changes are measured from the same initialization. All global statistics use unique model coordinates, excluding vocabulary padding and duplicate entries for tied weights. Per-layer statistics and absolute-threshold sensitivities are retained in the experiment archive. For an absolute threshold τ , sparsity is the fraction of coordinates with magnitude at most τ ; the figures show the fraction exceeding τ . Energy concentration uses squared Euclidean norm and ranks coordinates by absolute change.

For checkpoint-based measurements, the exported checkpoint contains FP32 master values, BF16 model values, first and second moments, optimizer hyperparameters, and step counters. Student and teacher values are verified against their saved models in the same precision. Forward and backward passes evaluate these values in FP32 with TF32 disabled. All proposed steps are computed from the same saved state with a non-fused Adam implementation. The online training optimizer uses its own fused implementation.

We compare three Adam states at the same checkpoint. *Saved* retains both moments and the step counter. *Reset m* zeros the first moment while retaining the second moment and step counter. *Fresh* zeros both moments and resets the step counter. Global norm clipping at 1 is applied before the Adam calculation, and weight decay is zero. The plotted controls use two cached response batches; the zero-gradient control is evaluated once. For the zero-gradient branch, the saved first moment continues to decay and move the master weights. Casting the saved FP32 master weights to BF16 without an optimizer step exactly reproduces the saved model. Figure 7 shows how the threshold changes the measured fraction of updated coordinates.

Setting the current gradient to zero while retaining the trained Adam moments changes 0.1403% of BF16 coordinates, compared with 0.1486% for PG. Resetting the first moment gives 0.0504% for PG; resetting both moments and the step counter gives 0.5140%. The parameters changed by Adam therefore depend on its accumulated moments and BF16 rounding, not just on which entries of the current gradient are nonzero. These controls supplement the update-sparsity measurements in Sections 4 and 5.

Figure 5(b,c) displays these BF16 controls. The full FP32 and BF16 measurements are retained in the experiment archive.

C.2 CACHED RESPONSES, GRADIENT OVERLAP, AND GRADIENT NORMS

These raw-gradient diagnostics use the M-PG/100 student, whereas Figure 3(b,c) measures teacher-conditioned optimizer steps at M-PG/500. Batches use seeds 42 and 43, one response per domain per batch, a 2,048-token prompt cap, and a 256-token generation cap. All eight responses reach the cap. Four response positions are sampled without replacement per response. Each response’s four prefixes receive equal weight. For domain-specific inputs, each teacher scores the response from its own domain; for shared inputs, each teacher scores all four responses with equal domain weights.

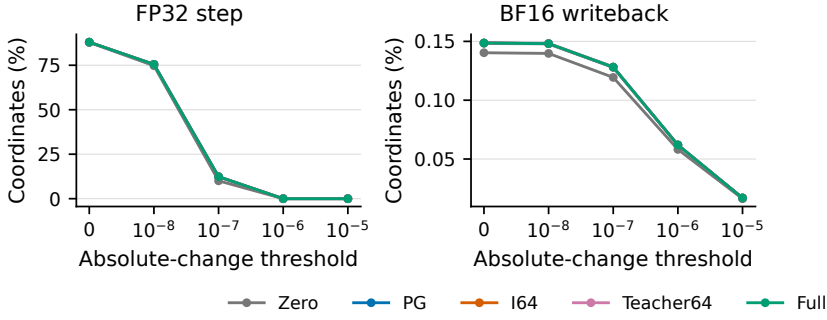


Figure 7: **Update sparsity depends on the numerical threshold.** FP32 steps and BF16 parameter changes; lines show batch means and shading their range.

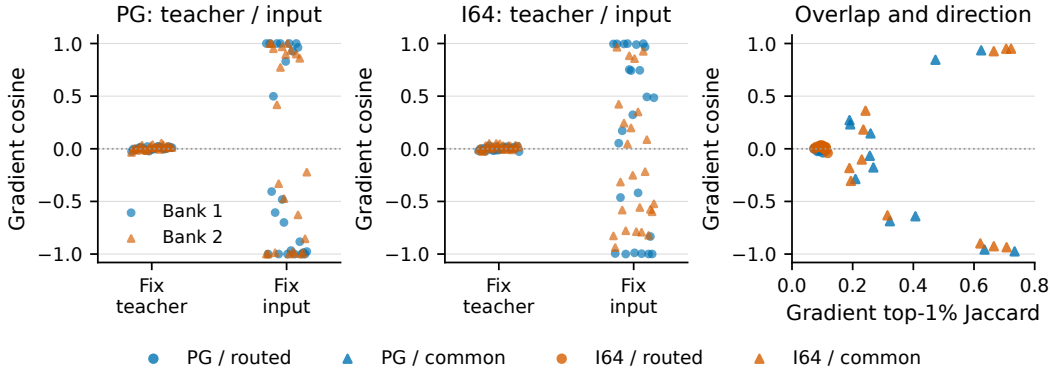


Figure 8: **Gradient overlap depends strongly on the input domain.** Left/center: teacher-by-input-domain controls. Right: top-1% gradient Jaccard versus cosine on domain-specific or shared inputs.

A third control evaluates every teacher on every input domain separately, keeping four prefixes per gradient. These conditions reuse the same responses, allowing paired comparisons within each batch.

Top- k coordinate sets rank absolute gradient values, using a stable coordinate index to break cutoff ties. Table 7 reports PG gradient norms for domain-specific inputs; the first batch’s instruction-following gradient has the reported norm 6.13×10^{-9} . Cosine similarity divides out the gradient norms, so a negative cosine does not imply a large opposing contribution to the aggregate raw gradient. Jaccard overlap measures selected locations and cosine measures signed direction. For reference, independent uniform sets of fraction ρ have expected-intersection to expected-union ratio $\rho/(2 - \rho)$, or approximately 0.00503 at $\rho = 1\%$. This reference uses uniform coordinates; layer and weight-magnitude structure remain in the measured gradients.

Applying Equation 8 to the largest-magnitude 1% of raw-gradient coordinates gives mean Jaccard overlaps of 0.0951 for PG and 0.0999 for I64 on domain-specific inputs, averaging the two batches and six teacher pairs. Their ranges are 0.0744–0.1077 and 0.0737–0.1178. When every teacher instead scores all four domain responses, the respective means are 0.3808 and 0.4577. The latter comparison changes both input composition and prefix count; the teacher-by-input-domain comparison below fixes the prefix count.

Holding the teacher fixed while changing the input domain gives mean gradient cosines of 0.0059 for PG and 0.0072 for I64. Holding the input domain fixed while changing teachers produces both aligned and opposed gradients. For domain-specific inputs the mean cosines are -0.0020 and 0.0081 , respectively, despite partial coordinate overlap. Figure 8 shows how input choice affects the teacher-overlap measurements in Section 4.

Table 6: **Vocabulary supervision produces similar single-step update concentration.** M-PG/100 means over four cached batches. NZ: nonzero coordinates (%); top-1%: squared step norm in the largest 1% of coordinates (%).

Loss	FP32 NZ	BF16 NZ	FP32 top-1%	BF16 top-1%
PG	87.982	0.149	8.845	100.000
I64	87.989	0.150	8.827	100.000
T64	87.989	0.150	8.828	100.000
Full	87.999	0.150	8.828	100.000

D VOCABULARY SUPERVISION AND COVERAGE

Single-step comparison using four response batches. The M-PG/100 row of Figure 4(a) uses four cached response batches with seeds 2042–2045. Each batch contains one response per domain. Generation uses the training prompt format, a 2,048-token prompt cap, and a 4,096-token response cap. The 16 response lengths range from 81 to 4,096 tokens; two responses reach the cap. Six positions per response give 96 prefixes. PG actions are independently sampled from the student at these fixed prefixes. All losses use the same prefixes, equal domain weights, and equal response weights within each domain. Gradients are globally clipped at norm 1 before applying the same saved Adam state. Table 6 adds the FP32 and energy-concentration measurements for this state.

The loss and Adam implementations match the two-batch controls in Appendix C. Appendix H.3 reports the actual I64 intersection coverage at all four student checkpoints.

Implementation details. For I64 (Equation 4), candidate sets and teacher scores are fixed at roll-out. Each student forward recomputes current probabilities at the saved candidate positions and refreshes the detached coefficients. PG also refreshes its detached log-ratio coefficients and uses no advantage or policy-ratio clipping.

Teacher-top-64 reference objective. The teacher-top-64 diagnostic uses the fixed teacher set T and full-vocabulary probabilities:

$$\ell_{T64} = \sum_{v \in T} \left[p_{\theta}(v) \log \frac{p_{\theta}(v)}{q_d(v)} - p_{\theta}(v) + q_d(v) \right]. \quad (12)$$

Its gradient is $\sum_{v \in T} p_{\theta}(v) \log(p_{\theta}(v)/q_d(v)) \nabla_{\theta} \log p_{\theta}(v)$. I64 instead uses the detached weights in Equation 4, normalized over the original student set S . T64 and full-vocabulary KL are used only for single-step comparisons. End-to-end training compares PG, I16, and I64; the I16 runs have no corresponding single-step gradient or update measurements.

For a fixed prefix, differentiating Equation 2 gives

$$\nabla_{\theta} \ell_{\text{full}} = \sum_{v \in \mathcal{V}} p_{\theta}(v) \left(\log \frac{p_{\theta}(v)}{q_d(v)} + 1 \right) \nabla_{\theta} \log p_{\theta}(v) \quad (13)$$

$$= \mathbb{E}_{a \sim p_{\theta}} \left[\log \frac{p_{\theta}(a)}{q_d(a)} \nabla_{\theta} \log p_{\theta}(a) \right], \quad (14)$$

using $\sum_v \nabla_{\theta} p_{\theta}(v) = 0$. This identity conditions on fixed prefixes and weights; the training loss additionally depends on the sampled response and the token weights induced by its final length and the normalization rule. Table 7 reports candidate-set coverage for the two-batch gradient-overlap and Adam-state controls.

E TEACHER DISTRIBUTION DISTANCE

We measure teacher distribution differences using Jensen–Shannon divergence on the full next-token vocabulary at identical prefixes:

$$\text{JS}(q_d, q_e) = \frac{1}{2} D_{\text{KL}}(q_d \| m) + \frac{1}{2} D_{\text{KL}}(q_e \| m), \quad m = \frac{1}{2}(q_d + q_e). \quad (15)$$

Table 7: **The teacher top-64 set captures nearly all student probability mass in these controls.** Means over sampled prefixes; gradient norms use domain-specific inputs. Entropy is in nats, and 100% may reflect rounding.

Bank	Domain	Mass in T (%)	Mass in I (%)	Entropy	$\ g_{PG}\ _2$
1	Math	100.0000	100.0000	0.3243	132.6
1	Code	99.9968	99.9965	0.7701	163.8
1	IF	100.0000	100.0000	1.85×10^{-5}	6.13×10^{-9}
1	Science	99.9184	99.9162	0.7405	54.82
2	Math	99.9991	99.9991	0.3221	2.605
2	Code	100.0000	100.0000	0.2962	3.167
2	IF	100.0000	100.0000	0.5004	55.52
2	Science	99.9993	99.9992	0.2442	0.8505

We average over the same equally weighted domains and prefixes for every pair. Figure 11(b) pairs this distance with the largest-magnitude coordinates of optimizer steps on shared inputs at M-PG/500. The earlier M-PG/100 raw-gradient diagnostic contains twelve batch-teacher-pair JS measurements, ranging from 0.00151 to 0.02714 nats; the per-pair measurements are retained in the experiment archive. Both diagnostics share teachers and prefixes within each batch, so teacher-pair observations are dependent.

F SUPPORTING MEASUREMENTS FOR THE MAIN FIGURES

Training-seed uncertainty. Figures 1(c–f) and 9 report means and sample standard deviations for three training seeds (42, 43, and 44). The expanded endpoint table and Figure 4(c) use seed 42; they do not estimate training-seed variability. In Figure 1(a), legend values are mean domain token shares over the plotted batches.

Teacher-update controls. Figure 3(a) compares cumulative parameter changes under DR at update 500; the heatmap uses a square-root color scale. Panels (b,c) start from the same M-PG/500 student parameters and Adam moments. With domain-specific inputs, each teacher scores responses from its own domain; with shared inputs, all teachers score the same prefixes. Each of four cached response batches contains six teacher pairs. Small points summarize pairs within a batch, and large points average batches. Equation 9 subtracts the Adam step with the current gradient set to zero before computing FP32 update cosines. It subtracts the zero-gradient baseline; the remaining difference still depends nonlinearly on the saved moments. SGD and reset- m steps are individually rescaled to the norm of the Adam step with trained moments; reset- m retains the second moment and step counter. Teacher pairs reuse teachers and responses, so pairwise observations are dependent. These calculations use non-fused PyTorch updates on copies of saved checkpoints and need not reproduce the fused training optimizer bit for bit. Mathematics-only and joint training also differ in mathematics prompt exposure (Appendix A).

Vocabulary-supervision controls. Figure 4(a,b) uses four cached response batches at each of four student checkpoints: M-PG and M-I64-DR at updates 100 and 500. Losses use identical prefixes and Adam moments within each checkpoint, while responses can differ between checkpoints. Panel (a) reports batch means. In panel (b), points are individual batches, lines connect PG batch means, and shading spans the I64 measurements. Direct pairwise comparisons of T64 update directions were not recorded; I16 appears only in the end-to-end training results. Panel (c) compares S-I64, M-I64, and M-I16 with PG at the same task, teacher setting, and update. Table 6 adds FP32 and energy-concentration measurements for the M-PG/100 comparison in Appendix D.

F.1 ADDITIONAL OPTIMIZER AND SAMPLING CONTROLS

Figure 2 uses the six cached response batches and training clipping records described in Appendix B.1. In panel (a), boxes mark the interquartile range and whiskers the 5th–95th percentiles; labels give the fraction of training updates clipped. Reset- m and reset- (m, v) retain the optimizer

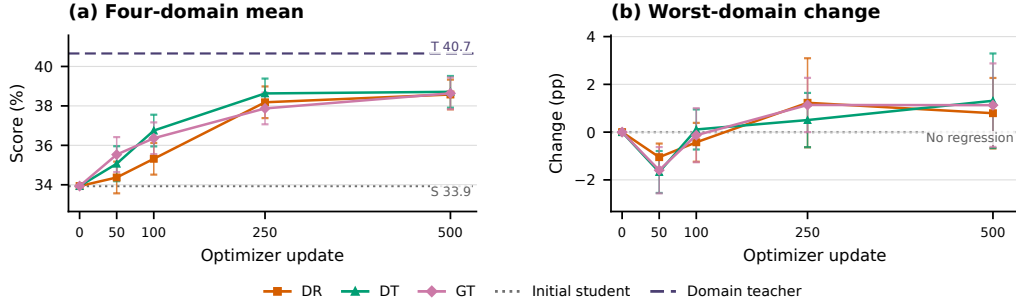


Figure 9: **Mean performance gains can coexist with a regression in an individual domain.** (a) Four-domain mean; S/T mark the initial student/teacher. (b) Worst-domain change from initialization; zero means no regression. Error bars show mean $\pm$ SD over three training seeds.

step counter. Panel (c) averages four domain probe losses within each batch; error bars show sample standard deviations over the three batches at each checkpoint.

Figure 5(a) uses three M-PG/100 response batches, with a 4,096-token response cap, containing 48 responses and 288 prefixes. Twelve fixed subsets of prefixes each receive two independent draws of 1, 16, or 64 PG actions per prefix. I64 is evaluated once per subset. Increasing the sample count raises the mean PG gradient cosine to full-vocabulary KL from 0.5374 to 0.8652 and 0.9667; I64 reaches 0.999988. These repeats measure sampling variation at a fixed checkpoint. Panels (b,c) use the separate two-batch experiment with a 256-token response cap (Appendix C); its zero-gradient control is evaluated once. Reset- m retains the second moment and step counter, whereas fresh Adam resets both moments and the counter. The sampling and moment-reset experiments use different response batches and are not paired.

Cumulative parameter changes. Figure 10 shows the trajectories underlying the update-500 summary in Figure 3(a).

G PROMPT EXPOSURE AND CANDIDATE-SET WEIGHTS

When each domain loss is averaged separately, its prompt quota and loss weight control different quantities. Increasing the quota at fixed λ_d changes the number of sampled responses and the variance of the gradient estimate, without directly increasing the domain’s expected contribution to the total gradient. Mathematics-only runs receive four times as many mathematics prompts per update as joint runs. Their score differences therefore cannot be attributed to teacher count alone.

Candidate-set coverage also scales the distillation gradient. For the student candidate set S and its intersection I with the teacher set in Equation 4, the retained student weights sum to $r(h) = p_\theta(I | h)/p_\theta(S | h)$. At a fixed prefix with a nonempty intersection and detached coefficients, the I64 gradient equals $r(h)$ times the gradient obtained by renormalizing the same weights over I . Equal response weights therefore do not imply equal gradient scale at every prefix. Measuring the retained mass and empty-intersection rate by domain can reveal unequal coverage under a fixed candidate size. Enlarging the candidate set or renormalizing its intersection changes the objective and requires a separate performance comparison.

H ADDITIONAL TEST RESULTS AND ROBUSTNESS CHECKS

H.1 ADDITIONAL TEST SETS FOR THE SAME FOUR DOMAINS

The ten seed-42 joint-training trajectories use the development benchmarks and checkpoint grid in Appendix A. The following analyses evaluate selected checkpoints on separate test questions, measure candidate-set coverage, and check sensitivity to known training–evaluation overlap.

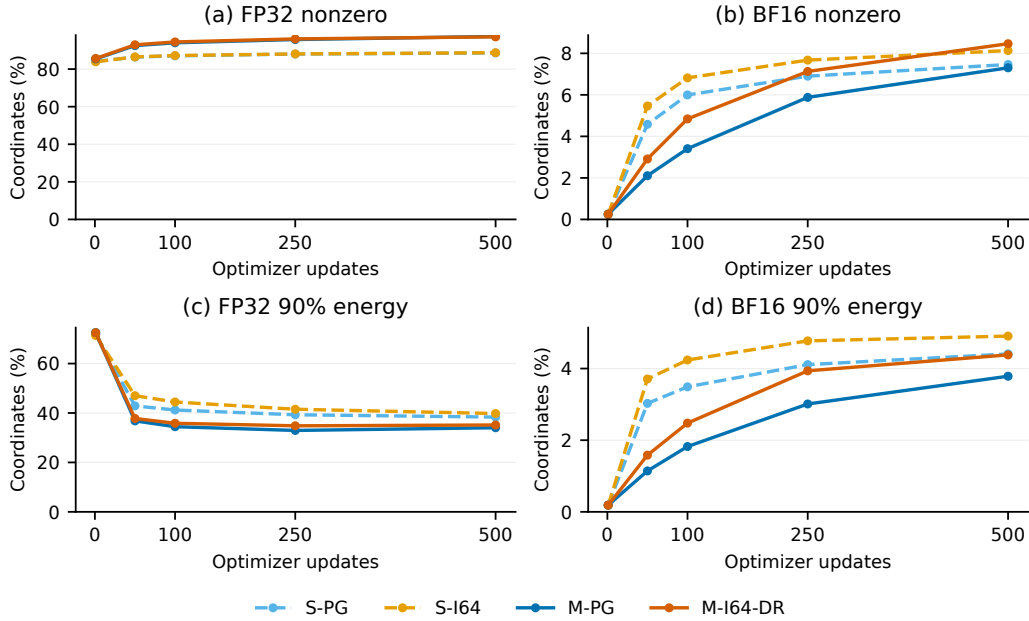


Figure 10: **Cumulative BF16 changes remain much sparser than FP32 changes.** PG/I64 under DR in single-teacher (S) and joint (M) training: (a,b) nonzero fractions; (c,d) coordinate fractions containing 90% of squared change. Lines connect measured updates 1, 50, 100, 250, and 500.

Table 8 uses additional test sets fixed before inference, with 256 questions per domain: 1,024 questions and 1,792 responses per model. Mathematics and code come from the Skywork-ORL-RL-Data mathematics and TACO-based code pools; instruction following uses Nemotron-RL-instruction_following; science uses the Nemotron-RL-knowledge-mcqa validation split restricted by a fixed natural-science keyword filter. Questions are the first eligible unique candidates in the fixed source order, selected without model scores. Normalized exact-match and lexical near-duplicate checks remove questions matching the original training bundle or known evaluation pools. These checks do not establish exhaustive semantic decontamination or rule out pretraining exposure.

Scoring uses mathematics answer verification, code unit tests, instruction-following constraints, and science multiple-choice accuracy. Mathematics, code, and instruction following use greedy decoding; science uses four responses per question at temperature 1. The context budget is 32,768 tokens, with thinking disabled. The data distributions and code/instruction-following scorers differ from the development benchmarks, so their scores are reported separately. The four-domain mean weights domains equally. Question-paired bootstrap intervals quantify finite-test uncertainty and do not estimate variation from training seeds.

H.2 CHECKPOINT SELECTION BEFORE TEST EVALUATION

We apply the three selection rules in Section 6 to the development grid in Appendix A. Checkpoint choices are fixed before test inference and remain unchanged after the test and overlap-sensitivity analyses. Both development-based rules choose update 250 for PG-DT/Adam and update 500 for the other nine trajectories. The separate PG-DT/Adam update-250 test evaluation allows direct comparison with its update-500 endpoint.

The retention constraint does not change the checkpoint selected by maximum development mean for any trajectory. On the additional test sets, the update-250 PG-DT/Adam checkpoint scores 0.366 percentage points below its update-500 endpoint in four-domain mean. Thus earlier selection by development mean does not improve the separate-test mean in this case. All selected models satisfy the retention and useful-gain conditions when assessed using test point estimates, which is not a simultaneous confidence guarantee. PG-DR/Adam nevertheless has two development checkpoints

Table 8: **Method rankings depend on the task and test set.** Additional test-set scores (%) for the same four domains; mean weights domains equally.

Configuration	Update	Math	Code	IF	Science	Mean
Initial student	0	42.97	23.44	25.78	62.21	38.60
PG-DR / Adam	500	46.48	26.17	37.50	64.06	43.55
PG-DT / Adam	250	44.53	26.56	36.72	64.75	43.14
PG-DT / Adam	500	44.92	27.34	38.28	63.48	43.51
PG-GT / Adam	500	44.92	28.91	39.06	65.14	44.51
I16-DR / Adam	500	44.92	28.52	40.23	64.55	44.56
I64-DR / Adam	500	46.88	26.17	39.06	65.72	44.46
I64-DT / Adam	500	43.36	27.73	39.06	64.55	43.68
I64-GT / Adam	500	45.31	27.34	41.02	66.21	44.97
PG-DR / SGD	500	46.88	27.73	37.11	62.21	43.48
PG-DT / SGD	500	46.48	28.52	37.11	63.48	43.90
PG-GT / SGD	500	44.53	26.17	36.72	64.45	42.97

Table 9: **The retention constraint leaves every checkpoint choice unchanged.** Test Δ is the selected-checkpoint mean minus the update-500 mean (percentage points).

Trajectory	Endpoint	Max dev. mean	Retention	Test Δ (pp)
Other nine trajectories	500	500	500	0.000
PG-DT / Adam	500	250	250	-0.366

outside the retention tolerance: GPQA falls from 32.197% initially to 28.030% at update 50 and 30.177% at update 250, before reaching 34.975% at update 500. Evaluating only the final checkpoint would miss these intermediate regressions.

H.3 PROBABILITY MASS RETAINED BY THE CANDIDATE INTERSECTION

For each of four student checkpoints (M-PG and M-I64-DR at updates 100 and 500), four cached response batches provide one response per domain and six sampled positions per response: 96 prefixes per checkpoint. Each response is scored by its domain teacher. We measure $p_\theta(I | h)$ for the actual intersection $I = S \cap T$ and the retained fraction $r = p_\theta(I | h) / p_\theta(S | h)$.

The minimum student probability mass in the actual student–teacher top-64 intersection is 96.55% over the 384 sampled prefixes, and no intersection is empty (Table 10). Thus the single-step I64/full-vocabulary comparisons use prefixes where truncation discards little student probability mass. They do not establish the same behavior when the intersection has low probability mass, with other teachers, or during I16 training. The four batches repeat response sampling at each checkpoint; they are not independent training seeds, and responses may differ between checkpoints.

H.4 SENSITIVITY TO KNOWN MATHEMATICS OVERLAP

The fixed Qwen training pool contains five records matching four known MATH-500 problems, including variants in wording or answer format. The affected MATH test identifiers are counting-and-probability problem 430 and intermediate-algebra problems 1849, 2146, and 582. Their presence in the pool does not show whether a run sampled them or whether they improved its score. We remove these four questions from the existing per-question evaluation results, retaining the original 500-question results as the main protocol.

Removing the four questions leaves the four-domain-mean ranking unchanged for all twelve reported checkpoints. MATH-496 names the retained subset, not a new official benchmark. We keep the checkpoint choices fixed. This analysis does not retrain on a filtered pool or provide an exhaustive contamination audit.

Table 10: **The minimum observed I64 intersection mass is 96.55%.** Statistics over sampled prefixes; $p(I)$ is in percent and $r = p_\theta(I | h)/p_\theta(S | h)$.

Checkpoint	$p(I)$: mean	$p(I)$: min	$ I $: mean	$ I $: min	r : mean	r : min
M-PG/100	99.9141	96.5482	58.96	43	0.9999413	0.9977714
M-PG/500	99.9952	99.7825	61.59	50	0.9999975	0.9998511
M-I64-DR/100	99.9564	97.5664	60.08	36	0.9997331	0.9757993
M-I64-DR/500	99.9826	99.3722	61.75	51	0.9999912	0.9997005

Table 11: **Removing four overlapping mathematics questions leaves the mean-score ranking unchanged.** Mean (496) substitutes MATH-496 for MATH-500 and keeps the other domain scores fixed.

Configuration	Update	MATH-500	MATH-496	Mean (500)	Mean (496)
Initial student	0	68.80	69.15	33.93	34.02
PG-DR / Adam	500	73.20	73.79	38.29	38.43
PG-DT / Adam	250	77.00	77.42	38.78	38.88
PG-DT / Adam	500	73.20	73.59	38.03	38.13
PG-GT / Adam	500	76.20	76.81	38.36	38.52
I16-DR / Adam	500	75.40	76.01	38.65	38.80
I64-DR / Adam	500	75.80	76.41	38.58	38.73
I64-DT / Adam	500	74.00	74.40	38.71	38.81
I64-GT / Adam	500	74.20	74.60	38.64	38.74
PG-DR / SGD	500	76.40	77.02	39.46	39.62
PG-DT / SGD	500	75.20	75.81	39.14	39.29
PG-GT / SGD	500	75.00	75.60	38.39	38.55

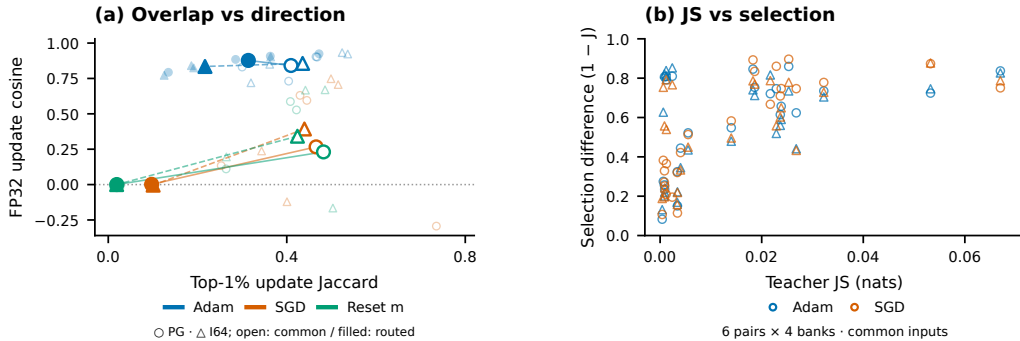


Figure 11: **Coordinate overlap and update direction measure different properties.** (a) Top-1% FP32 update-set Jaccard versus cosine; small/large points show batch/overall means. (b) Teacher-distribution JS divergence versus $1 - \text{Jaccard}$ on shared inputs; each point is a teacher pair within a batch.

H.5 TEACHER-UPDATE OVERLAP AND DISTRIBUTIONAL SIMILARITY

Figure 11 supplements the zero-gradient baseline subtraction at M-PG/500 using the controls in Appendix F. It compares PG and I64 under Adam with trained moments, reset- m , and norm-matched SGD. Shared inputs hold student prefixes fixed between teachers; domain-specific inputs change both the teacher and the input domain. Each of four batches contains six teacher pairs; these observations are dependent because they reuse teachers and responses. These gradient and update measurements do not directly measure changes in task performance.